

Senior Software Engineer, Data 2026-05-08

Guidelines

- Time: ~2 hours at home. Don't spend more and however far you get is fine.
- Stack: Python + your choice of database (SQLite is fine). Everything else is your call (libraries, schema, project structure).
- Submit by pushing to a GitHub repo and sharing the link with us.
- Include a brief README with any notes on your approach or what you'd do to handle this at scale.

Assignment

Commit 1

Attached is a denormalized CSV of historical stock data. Design a normalized database schema and build a Python pipeline that loads the data into it. The pipeline should be idempotent - running it multiple times, including on updated versions of the source CSV, should keep the target database in sync.

Include at least one example query against your normalized schema to show it's usable end-to-end. Anything that exercises a join and an aggregation works - cumulative return per company over the period is one option.

 stock-data-se-owl.csv

Commit 2

Once Commit 1 is in good shape, send a second commit handling the following change to the source data:

A new column `mktcap_usd` has been added to the source CSV. Existing rows should be backfilled where appropriate. In addition, Apple's historical close prices and volumes have been updated to reflect a recent 2-for-1 stock split adjustment — every Apple row now has different values than before. Re-run your pipeline against the new CSV; it should handle the schema migration *and* the in-place updates correctly.

We're looking at three things: the migration itself, the backfill strategy, and how the change propagates through your pipeline (queries, downstream tables, anything that depended on the

old shape). This kind of change happens frequently in real systems, so set up accordingly.

 stock-data-se-owl-v2.csv

Next Steps

After submission there will be an hour-long in-person session where you'll walk us through what you built.

What We Care About

How you prioritize and the decisions you make.

Reach out if anything is unclear.